

The Geometric Distribution

What is the Geometric Distribution?

The **geometric distribution** models the number of independent trials needed to achieve the first success in a sequence of Bernoulli trials (trials with only two outcomes: success with probability p or failure with probability $1 - p$).

Examples:

- Flipping a coin until you get heads
- Rolling a die until you get a 6
- Taking free throws until you make one

The probability mass function is:

$$P(X = n) = (1 - p)^{n-1} \cdot p$$

where n is the trial number of the first success.

Explaining the Last Step

The simplification uses the **infinite geometric series formula**:

Starting with:

$$\sum_{n=1}^{\infty} \binom{n-1}{0} 0.99^{n-1} \cdot 0.05 \cdot 0.95^{n-1}$$

Step 1: Note that $\binom{n-1}{0} = 1$, so:

$$= 0.05 \sum_{n=1}^{\infty} (0.99)^{n-1} \cdot (0.95)^{n-1}$$

Step 2: Combine the terms:

$$= 0.05 \sum_{n=1}^{\infty} (0.99 \cdot 0.95)^{n-1}$$

Step 3: This is a geometric series $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ where $r = 0.99 \cdot 0.95$:

$$= 0.05 \cdot \frac{1}{1 - 0.95 \cdot 0.99}$$

$$= \frac{0.05}{1 - 0.95 \cdot 0.99}$$

The key insight is recognizing that when you have a sum of the form $\sum ar^n$, you can use the geometric series formula to get a closed-form expression.